

Factors associated with acute respiratory illness in day care children.

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The aim of this study was to investigate the relationship between child characteristics, parental and environmental factors and the occurrence of acute respiratory illness (ARI) and acute otitis media (AOM) among Finnish children attending day care centres (DCCs). The study was a cross-sectional questionnaire of 594 children aged 1-6 y from 18 DCCs in Helsinki, Finland. Recurrent ($>$ or $=4$ diseases/y) ARI was present in 44% of the 1-3-y-olds and 23% of the 4-6-y-olds, and recurrent AOM in 15% and 2.5%, respectively. Parent atopic disease (odds ratio (OR) 1.53, $p = 0.033$), mother's academic education (OR 1.77, $p = 0.008$) and a medium length of DCC attendance compared to a short period (OR 1.67, $p = 0.049$) increased, while furry pets (OR 0.44, $p = 0.003$) and older child age (OR 0.38, $p < 0.001$) reduced the risk of recurrent ARI. Recurrent ARI (OR 3.96, $p = 0.008$), mother's academic education (OR 5.02, $p = 0.003$), and a medium length of DCC attendance compared to a short period (OR 3.34, $p = 0.044$) increased, while partial breastfeeding $>$ or $=6$ months (OR 0.20, $p = 0.002$) and older child age (OR 0.05, $p < 0.001$) reduced the risk of recurrent AOM. Parental and environmental factors had a significant impact on recurrent ARI and AOM episodes in children attending DCCs. These risk factors should be considered in future studies intending to reduce DCC infections.